

Where the Loops Come From

Why some networks branch and others join back up

Reading and note-making task • Year 9 • Chemistry, Biology & Geography

Part A Before you read: draw and predict

Do not read the essay yet.

A1. In the boxes below, sketch each network from memory. Do not worry about accuracy; worry about whether lines ever join back together.

A river and its streams	The veins in a leaf	The streets near school

A2. In which of your three sketches does a line ever leave a point and come back to it? Circle them.

A3. Building an extra connection always costs extra material. So why would anything ever build one? Your best guess:

Keep this page. You will mark your own prediction at the end.

Glossary strip use while reading

network	any set of points joined by links: streams, streets, tubes, roots
tree (in maths)	a network where no route ever returns to where it started
loop	a route that does return, making a ring in the network
slime mould	a single huge cell that spreads out and finds food without a brain
fungal network	threads growing through soil, connecting to tree roots
spare route	an extra link, not needed unless something else fails
predictable	roughly known in advance; tomorrow's job much like today's
outlet	the fixed place a river system finally drains to

Part B Cornell notes: five networks and one rule

Fill the **Notes** column as you read. Some cues are given; write your own in the blank cue lines.

Cue 1. FROST Why does a frost arm that pokes ahead grow even faster? What makes frost different from every other network here?	Notes
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Cue 2. THE COST OF A LOOP Why is a loop always more expensive than no loop? <i>Your own cue:</i> _____	Notes
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Cue 3. RIVERS What makes a river's job <i>fairly</i> predictable? Give two things. Name the exceptions the article admits to.	Notes
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Cue 4. SLIME MOULD Two steps in its method: spread, then ...? What three things were measured in the Tokyo comparison?	Notes
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Cue 5. THE RULE Complete both halves: predictable traffic → changing traffic → <i>Your own cue:</i> _____	Notes
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Cue 6. FUNGI, AND THE WARNING What is definitely true about fungal networks, and what is doubtful? <i>Your own cue:</i> _____	Notes
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Cue 7. CITIES Cities do have planners. So what is it that nobody does? <i>Your own cue:</i> _____	Notes
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Summary box. In no more than **40 words**, and without using the words *river*, *slime* or *city*, state the rule this essay is about.

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Part C Cloze: the argument in one paragraph

Fill each gap. Every answer is one or two words.

Frost branches, but nothing ever _____ along it, so its shape answers only to the water around it. Every other network here has a job, and that job costs material.

A network with no rings in it is called a _____. Adding a ring means adding a _____, which always costs more to build.

A river _____ builds one, because its water mainly goes one way, towards an outlet that stays put, and tomorrow's job is much like today's. When the job is that _____, a tree is usually enough.

A slime mould does build a few. It first _____ everywhere, then thickens the tubes carrying the most and lets the rest wither. It keeps spare links because it cannot know where the next _____ will appear.

So the rule is this: loops are worth their cost only when the traffic keeps _____, or when a link might get _____. Fungi and cities both face exactly that, which is why both of them are full of rings.

Part D Understanding and reflection

1. **Sorting task.** Predict whether each of these has loops or not, then say what its traffic is. *Lightning, the London Underground, the veins in a leaf, a tree's branches, blood capillaries, a drainage ditch system.*

Tree: no loops	Has loops

Pick the one you were least sure about and explain what made it difficult.

2. **Test the claim.** A friend says: "Loops are always better, so everything should have them." Using one example from the article, show that this is wrong, and state the condition under which loops stop being worth their cost.

3. **The careful bit.** The essay says the popular story about the wood-wide web is stronger than the evidence supports. Write down what is agreed and what is disputed, in your own words, in two lines.

4. **Push the idea further.** Think of a network built by people that changed from a tree into a loop, or the other way round, when the traffic on it changed. Describe it.

5. **Back to your prediction.** Reread your answer to A3. What did you get right, and what would you now change?

Part E At the screen

Use the interactive companion. Write what you observe, not what you expected.

1. **Frost.** Let it grow, then watch the “rings closed by accident” counter. It is usually zero but not always. What has to happen for a ring to appear, and does the frost gain anything from it?

2. **Rivers.** The loop counter here stays at zero no matter how many landscapes you generate. Explain why the rule the model follows makes a ring impossible.

3. **Shortest or safest.** Place seven towns. Record the total length with no spare links, then with three. Now cut links at random ten times in each case and count how often a town gets stranded.

	Total length	Towns stranded in 10 cuts
No spare links		
Three spare links		

What exactly did the extra length buy you?

4. **Slime mould.** Run it on “same delivery every time” and record the surviving loops. Reset, switch to “deliveries move around” and record again. The creature’s rule is identical in both runs. So what caused the difference?

Teacher key and notes

Part C cloze. travels; tree; link (accept “connection”); almost never; predictable; spreads; flake (accept “food”); changing; broken (accept “cut”).

Part D1 sorting. No loops: lightning, a tree’s branches, a drainage ditch system. Loops: the London Underground, the veins in a leaf, blood capillaries. The two productive difficulties are the leaf, which most students draw as a tree and which is in fact heavily looped, and lightning, which branches without carrying repeated traffic at all and so belongs with frost rather than with rivers. A tree’s branches are the cleanest tree on the list, since sap moves one way to a known set of destinations.

Part D2. The intended answer is the river: loops there would add length while adding nothing, because the direction of flow and the position of the outlet do not change. The condition to state is that loops stop paying when the traffic is predictable and the routes are reliable. Watch for students who answer “because rivers are not alive”, which is true but is not the reason.

Part D4. Strong answers include a railway that gained a loop line as commuting patterns spread, an electricity grid moving from radial to meshed supply, or a village lane network absorbed into a ring road. Accept anything where the change in traffic is stated, not just the change in shape.

Part E. Expected figures: three spare links add roughly a fifth to the total length on seven towns and reduce strandings from most cuts to almost none. In the slime-mould module the fixed-delivery run collapses to a single path with zero loops, which looks like a broken simulation and is not; warn the class in advance. The moving-delivery run typically leaves two to five loops standing. The point students should reach unaided is that only the uncertainty changed between the two runs.

Pitching note. This edition matches the Year 9 version of the essay. Cue boxes 1–7 follow its sections in order. Section 5, the rule, is the hinge; if time is short, cut the fungal section rather than that one, since cities alone are enough to complete the argument. The fungal section is the only place where students meet a contested claim, and it repays being read aloud together.

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